Questions for review session
Instructor: Abolfazl Babaei

1) Write a piece of Python code that computes an approximation to sinh(x) using the equation given below.

$$\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$$

This code is given the values of x and tolerance as floats. Add terms to the summation as long as the absolute value of the current term is greater than the tolerance. Count the number of terms added to the sum. Return the sum of the series as the approximate value of sinh(x) followed by the number of terms added to the sum.

Note: It is not mandatory to write a function for this question but if you want you can answer this question by completing the function below.

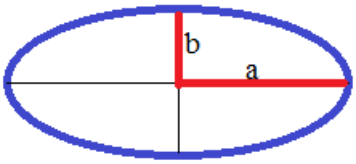
```
def Sinh (x, tolerance):
```

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2) Complete the Python code that calculates the X and Y coordinates of the points on an oval. The code is given the semi-major and semi-minor diameters of the oval (a, and b), and the number of intervals between equally spaced X coordinates. Create a list of equally spaced X coordinates from x = 0 to x = 2*a based on the number of intervals. Using the equally spaced X coordinates and equation 1 create a list of Y coordinates, one Y coordinate for each X coordinate. Print a table that contains the X coordinates in one column and the corresponding Y coordinates in the other column.

Hint: The equation below should be used to answer this question. In order to find Y coordinates, you need to initialize X coordinates, a, and b and find corresponding values of Y coordinates.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$




where a and b are the semi-major and semi-minor of the oval and x and y are the coordinates of the points on the oval.

Note: You can write the import statement(s) required by this function here and complete the function. However, answering this question in the format of a function is not mandatory.

```
def computeCoords(radius, intervals):
```

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3) Write a piece of code in Python that grabs N in the input and print out the N number of the sequence below: (in this example, the user had entered 10 as the number of the desired terms)

```
1
1, 5
1, 5, 12
1, 5, 12, 22
1, 5, 12, 22, 35
1, 5, 12, 22, 35, 51
1, 5, 12, 22, 35, 51, 70
1, 5, 12, 22, 35, 51, 70, 92
1, 5, 12, 22, 35, 51, 70, 92, 117
1, 5, 12, 22, 35, 51, 70, 92, 117, 145
```

Finding the relationship between the terms is a part of the question mark.
Hint: In order to find the relationship between the terms of the series, you need to find how each term is built using the pervious term and its index. For example:

5=1+4
12=5+4+3
22=12+4+3+3
35=22+4+3+3+3
51=35+4+3+3+3+3
.
.
.
.